Lecture 1
Descriptive Regression with a Binary Covariate

August 24, 2023

Outline

What are descriptive statistics?

- Descriptive statistics are summaries of the data we have.
- The data we have might be a sample (e.g., an exit poll) or it might be the whole population (e.g., administrative vote counts), sometimes known as a census.
- Having the whole population is becoming more common in the era of Big Data.
- Note: if you are interested in the causal effects of a treatment (a drug, a policy, etc.), you can never really have the whole population (we will discuss this in a few weeks).

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What makes a good summary?

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From the google definition:

1. brief
2. includes main points

What makes a good descriptive statistic?

1. brief: at most a few numbers
2. includes main points:
   - sufficient to describe main characteristics of the data
     - there is a technical definition of sufficiency that I can discuss in office hours for those interested
   - minimizes the information lost by reducing the data
     - I'm using "information" in a loose sense here, will formalize a bit later
     - reducing the data means taking the data, which are often represented by $n * v$ numbers (with n the number of observations and v the number of variables) and reducing to a descriptive statistic that is represented by a few numbers.

Enough forest. Trees (examples) are coming!

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"Hold on, where's the forest again?"

It is easy in a statistics class to focus on the specifics. I want you to remember why we are doing what we are doing.

Outline

Univariate Example: SCOTUS data

- Data on 27 justices from the Warren (‘53 - ‘69), Burger (‘69 - ‘86), and Rehnquist (‘86 - ‘05) courts
- Data can be interpreted as a census for the 1953 - 2005 era.
- One Variable: Percentage of votes in liberal direction for each justice in civil liberties cases

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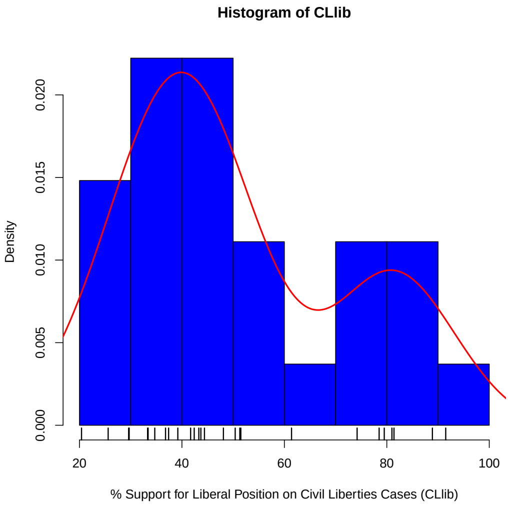
Describing Univariate Data



Description/Summarization

- Even for univariate data, we may want to focus on one or two characteristics of the population instead of describing it in its entirety.
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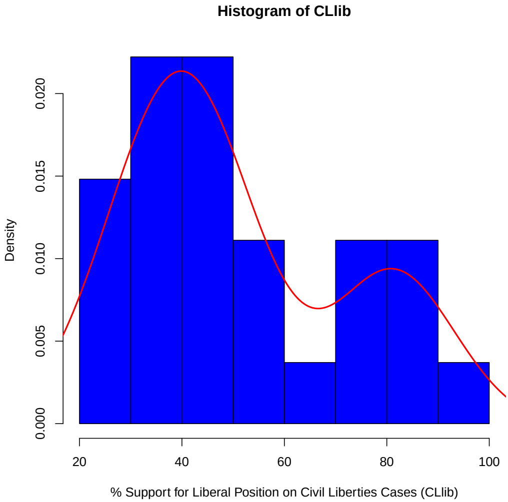
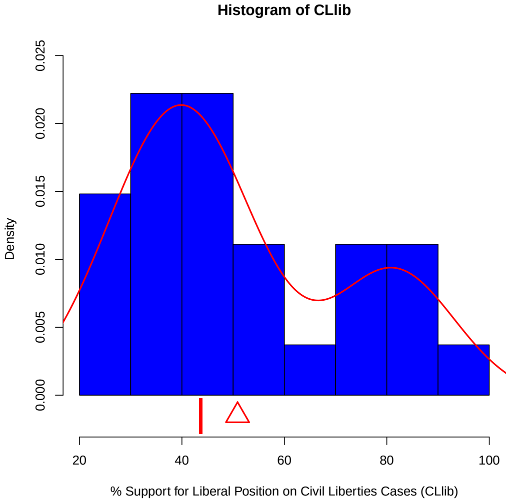
Univariate Activity



Find the rough location of the mean and median.

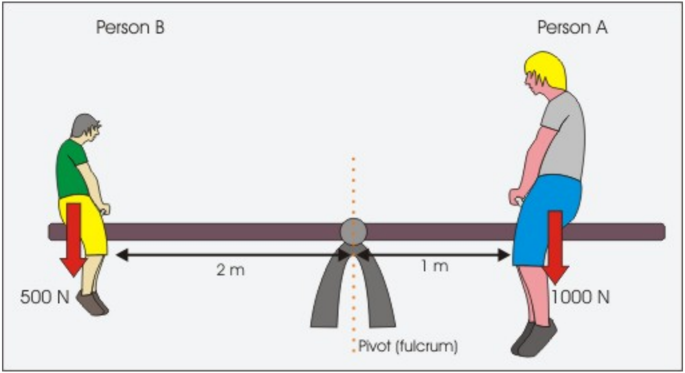
Univariate Activity Solution



Mean/Median Discussion

In pairs, discuss what the mean and median represent in the context of the CLib variable for these justices. What does it mean for the mean to be larger than the median in this context?

Mean as a balance point



Choosing a descriptive statistic

How might we choose between these descriptive statistics?

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A Least Squares Summary

Suppose we want to pick a single number (m) for the middle of the data that summarizes all the values for y, by minimizing the sum of squared residuals (i.e., least squares).

$$SSR(\tilde{m}) = \sum_{i=1}^{N} (y_i - \tilde{m})^2.$$

Activity: Using calculus, derive the least squares estimator

1. Calculate the derivative of SSR with respect to $\tilde{m}$
2. Set the derivative equal to 0
3. Solve for m

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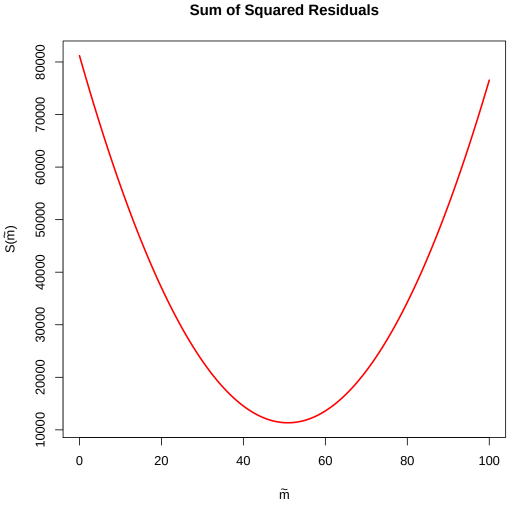
The Objective Function for SSR CLlib

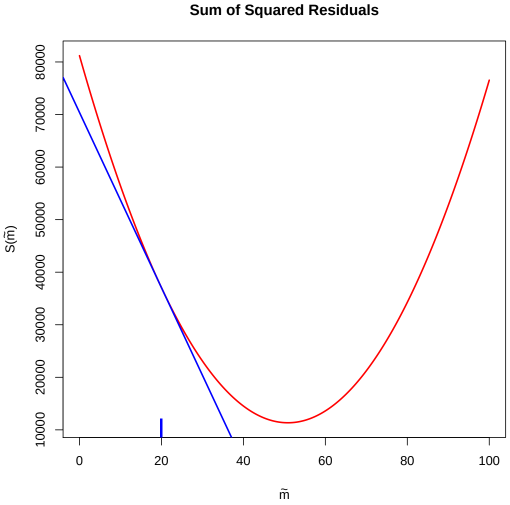


1.

$$SSR(\tilde{m}) = \sum_{i=1}^{N}(y_i - \tilde{m})^2$$
$$= \sum_{i=1}^{N}(y_i^2 - 2y_i\tilde{m} + \tilde{m}^2)$$

$$\frac{\partial SSR(\tilde{m})}{\partial \tilde{m}} = \sum_{i=1}^{N}(-2y_i + 2\tilde{m})$$

The Slope of the Tangent Line for SSR CLlib



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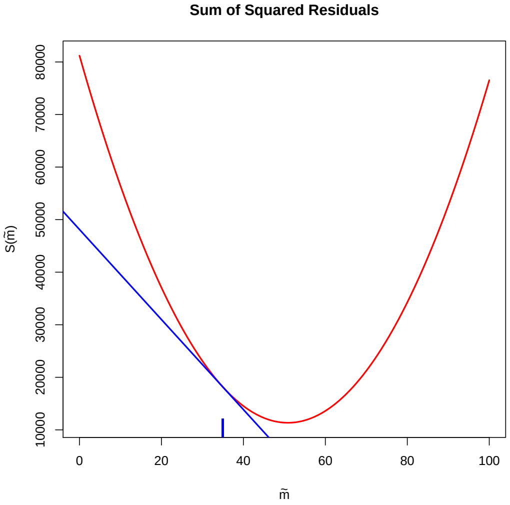

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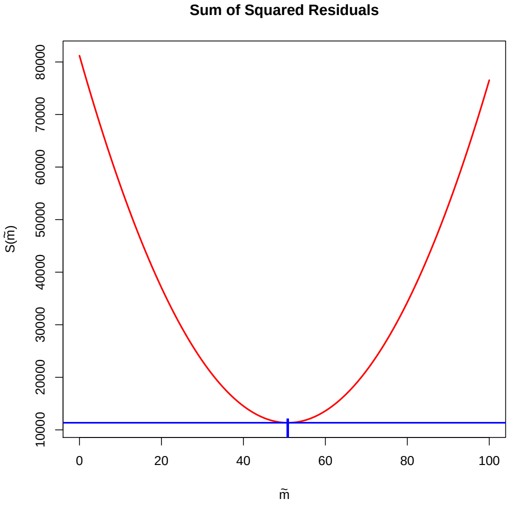

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$$\frac{\partial S(m)}{\partial \tilde{m}} = \sum_{i=1}^{N}(-2y_i + 2\tilde{m})$$

2.

$$0 = \sum_{i=1}^{N}(-2y_i + 2m)$$

3.

$$m = \frac{1}{N}\sum_{i=1}^{N} y_i$$

Therefore, the mean is a least squares summary.

A Least Absolute Value Summary

Suppose we want to pick a single number (m) for the middle of the data that summarizes all the values for y, by minimizing the sum of absolute deviations.

$$SAV(\tilde{m}) = \sum_{i=1}^{N} |y_i - \tilde{m}|.$$

- Why is this harder to solve?
- Any guesses?

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Why we focus on means

- Means often provide a “good” summary of the data (and it is relatively easy to tell when they provide bad summaries).
- The properties of means are relatively easy to describe.
- Randomized experiments only identify mean causal effects (as do regression and matching when the appropriate assumptions hold).

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Variance and Standard Deviation

- Mean is the least squares predictor in terms of SSR.
- How small is "least" (i.e., how much spread around the mean)?
- Variance and Standard Deviation are useful transformations of SSR for the mean.

Population Variance

$$S^2 = \frac{\sum_{i=1}^{N}(y_i - \bar{y})^2}{N-1} = \frac{SSR}{N-1}$$

Population Standard Deviation

$$S = \sqrt{S^2}$$

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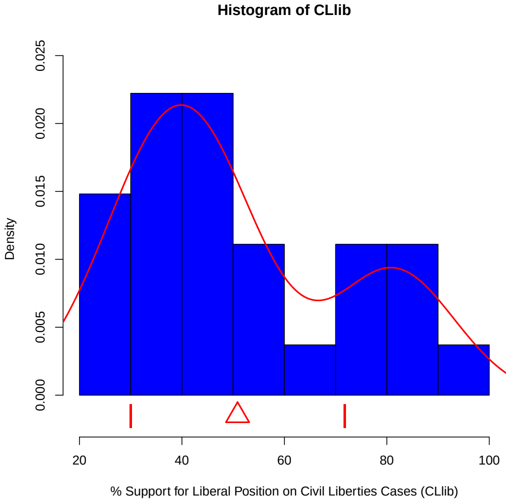
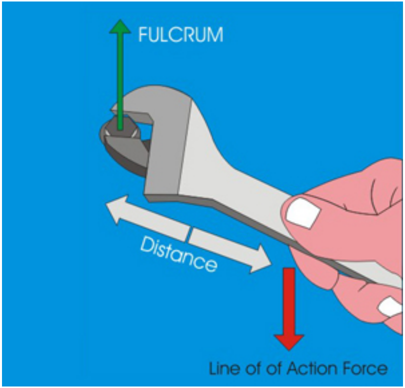


Variance as turning force



Why variance and standard deviation?

- Natural measures of accuracy for means.
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- For nonnormal data, means and variances provide much information via Chebyshev's inequality and other concentration inequalities.

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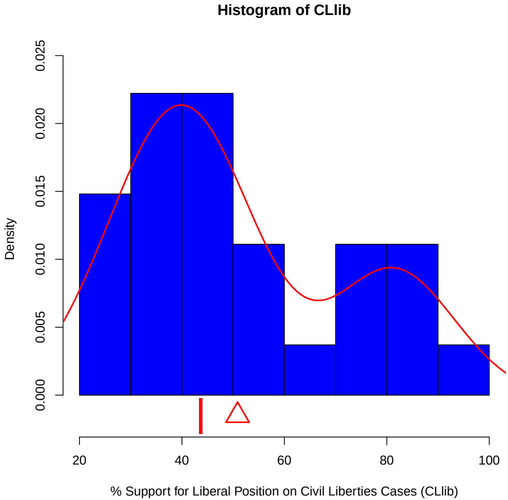
Outline

Conditional Means

- In this class, we focus on a single “outcome” variable.
- All additional variables adjust how we consider the outcome, often by conditioning.
- Typically, we consider the mean of y conditional on values of other variables.

From Univariate to Regression



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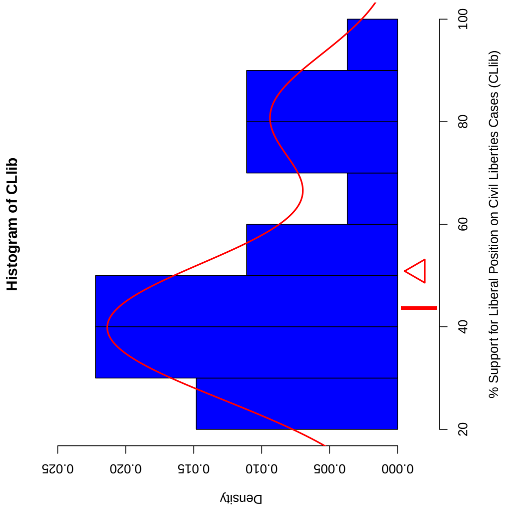
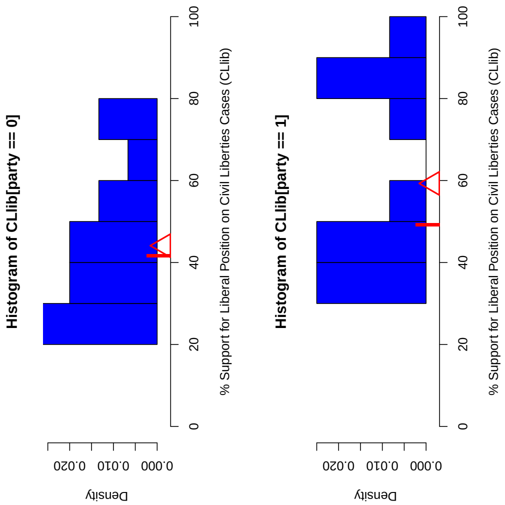


Binary Covariate Regression

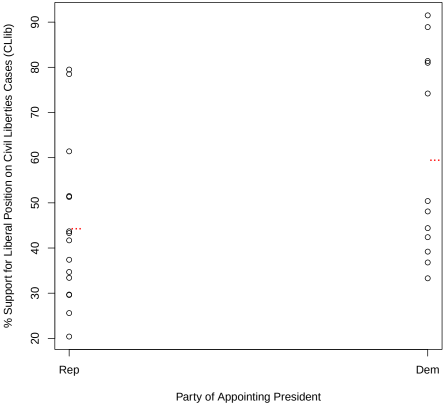


Conditional Mean/Median Discussion

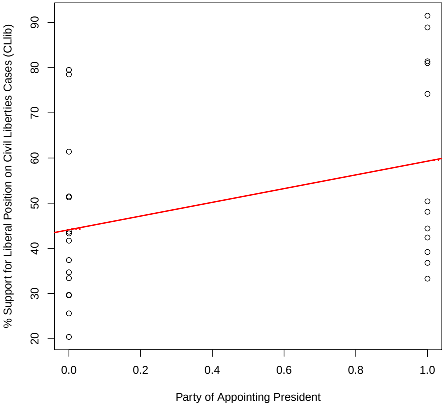
In pairs, discuss ...

1. What do the mean and median represent in the context of the CLib variable for the justices of each party?
2. What does it mean for the mean to be larger for the Democrats than the Republicans?
3. What does it mean for the difference in means to be larger than the difference in medians?

Scatterplot Representation



OLS Regression



What do the intercept and slope of the line represent?

Conditional Standard Deviations

Special Case: Binary Outcomes